A study on robust feature representations for grain density estimates in austenitic steel

Filip Ilic[1], Marc Masana[1,2], Lea Bogensperger[1], Harald Ganster[3], Thomas Pock[1]

[1] Graz University of Technology, Austria

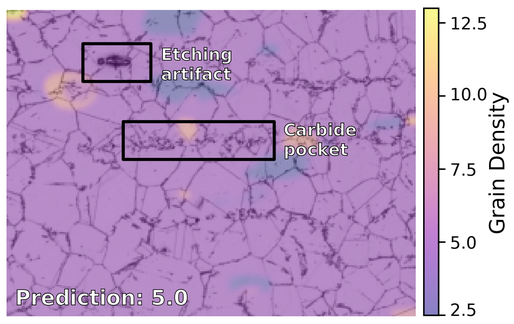
[2] Silicon Austria Labs, TU-Graz SAL DES Lab, Austria

[3] Joanneum Research Forschungsgesellschaft mbH, Austria

Problem Statement

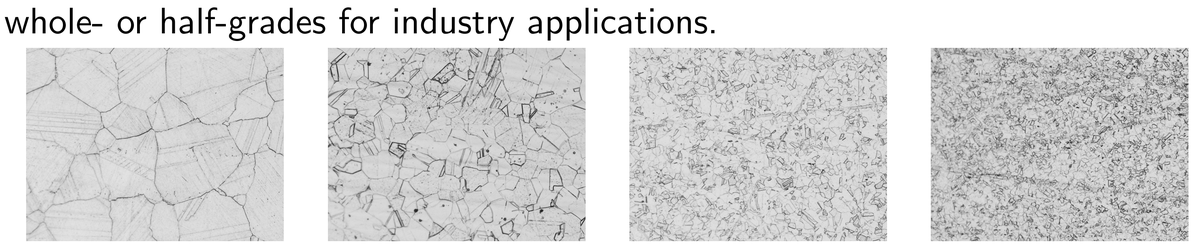
Structural properties of steel are determined in part by the grain density

- Goal: determine the grain density of austenitic steel samples
- Compare classification and regression networks on Out-of-Distribution data



Dataset

Grain density while fundamentally a continuum, it is often discretized to whole- or half-grades for industry applications.



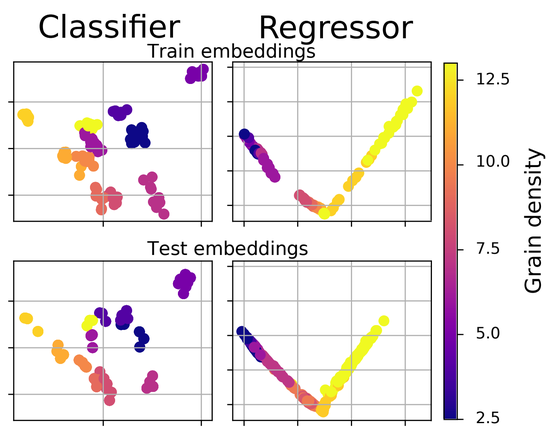
- The grain density increases from left to right and exhibits fractal-like self similarity at different scales
- Our new datasets, with samples captured at 100-fold magnification, consist of 242 images that have a resolution of 1280×960 - split into 125 train, 53 validation, and 64 test images
- Class distribution is kept balanced over all splits, totaling 11 classes

Methodology and Results

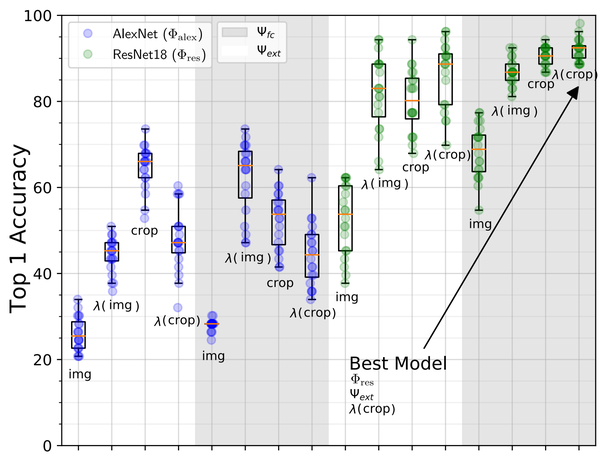
We focus on exploring the difference exhibited by regression and classification networks given different network structures and augmentation strategies.

	Classification		Regression	
	Crop	Image	Crop	Image
$\Phi_{\text{res}}, \Psi_{\text{ext}}$	89.91 ± 3.02	98.11 ± 0.09	86.42 ± 7.66	81.98 ± 8.80
$\Phi_{\text{res}}, \Psi_{\text{fc}}$	93.02 ± 4.07	97.92 ± 0.58	91.89 ± 2.68	86.98 ± 2.99

- We visualize the resulting feature space with PCA
- As seen in the table, classification networks yield better performance w.r.t. classification error
- The resulting feature space of classifiers and regressors differs; regressors appear to have a more robust feature representation



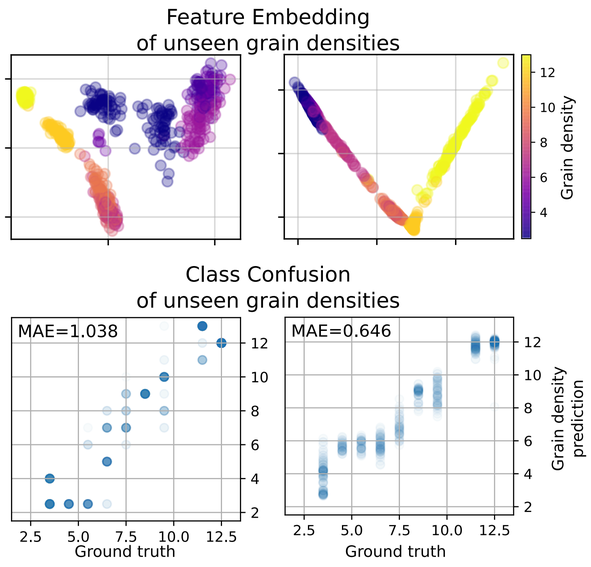
Ablation of Regressors



- Residual blocks show improved performance
- Proposed data augmentation technique (λ) yields performance increase
- Using image crops performs better than regression based on the whole image at once

Discussion

Out-of-Distribution Robustness



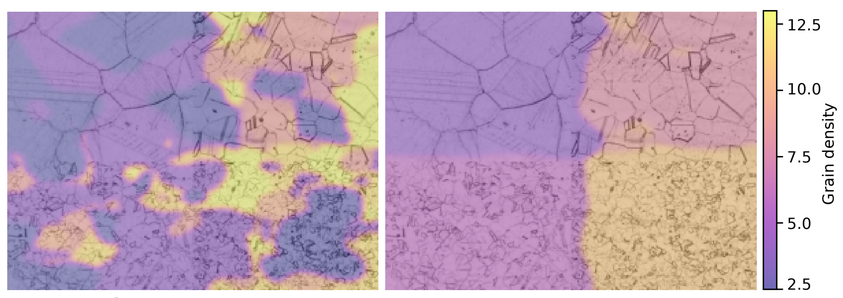
- Classifiers do not seem to encode the grain density as the most prominent dimension
- Regressors do encode the grain density in a linear fashion in the projection of the feature space
- The resulting error on OOD samples is shown in the confusion plots, which corresponds to the table below

We can also measure the advantage of an ordered feature space representation by probing the model with samples that deviate from the initial dataset, i.e. grain density samples that have not appeared in the training dataset.

Model	Train	Test	Unseen Samples
Best Classifier $(\Phi_{\text{res}}, \Psi_{\text{ext}}^{cls})$	0.000	0.019	1.038
Best Regressor $(\Phi_{\text{res}}, \Psi_{\text{fc}}^{reg})$	0.135	0.216	**0.646**

Inhomogeneity detection

We propose a small modification to the GradCAM algorithm to visualize regions that correspond to a class. The test image shown below is artificially spliced together from 4 images with different grain densities. This is possible because our classes can be ordered by grain density in a meaningful way.



$$L^* = \arg\max_c(L)$$

$$L^c = \text{ReLU}(\sum_k w_k^c A^k)$$

$$w_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

with A^k being the activation maps at the layer of inspection.

- Left Image: Classification network, trained on whole images
- Right Image: Classification network, trained on image patches

Note that the patch based model yields a clear seperation between the spliced images, whereas the network trained on whole images is unable to do so.

Conclusion

- Classification models yield overall better results than comparable regression models
- Regression models, however, encode the grain density quite clearly as the main axis in the learned feature space
- Due to this ordered feature space, regressors are more robust to Out-of-Distribution samples
- This shows that perhaps in production environments where OOD samples might be encountered more frequently choosing a model not only on the 'lab-performance' but also on its interpretable feature space representation makes sense

This work was supported by Land Steiermark *within the research initiative "Digital Material Valley Styria".*